



Module 1 Day 3: Surveying the Cyberspace

Class Instruction

Overview

Today's class focuses on professional pathways. You will get an overview of cybersecurity certifications, domains, and professions, and an in-depth review of the Security+ (Sec+) exam.

Class Objectives

By the end of class, you will be able to:

- Consider roles and career pathways within the cybersecurity space.
- Explain the landscape of certifications available to security professionals.
- Explore what the Security+ exam is and which InfoSec pathways benefit from the certification.

01. Welcome and Today's Themes

Welcome to the third day of class. We've only just scratched the surface of cybersecurity, but we have covered *a lot*. Today, we'll continue to expand this scope by examining various cybersecurity domains, careers, and certifications.

- You shouldn't feel pressured to know exactly what career path you want to pursue after this boot camp. Once you are exposed to the various skills, concepts, and domains covered in the various units of this course, you will have

a better idea. For now, we'll start by discussing the high-level domains that exist in the cybersecurity industry.

- You will also have an opportunity to search various job platforms to get a sense of the current cybersecurity hiring landscape.
 - Next, we will do a high-level overview of the Security+ Exam
 - Finally, at the end of class, you will participate in a fireside chat, which will provide you with an opportunity to hear about the careers and backgrounds of you and the TA, as well as advice on how to succeed in the class and in the cyber industry.
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02. Quick Review

- In the last class, we discussed how nearly every single node of a web application—the webpage, server, database, and even the users and administrators—can be targeted by a cybercriminal.
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03. Cybersecurity Domains

Consider this quote on entering the cybersecurity field from Tao Security's Richard Bejtlich:

- *"Providing advice on 'getting started in digital security' is similar to providing advice on 'getting started in medicine.' If you ask a neurosurgeon he or she may propose some sort of experiment with dead frog legs and batteries. If you ask a dermatologist you might get advice on protection from the sun whenever you go outside. Asking a 'security person' will likewise result in many different responses, depending on the individual's background and tastes" ([Bejtlich, 2017](#)).*

The field of cybersecurity is so vast that it is often hard to get straightforward advice on how to enter the industry. This is because the advice of each professional will vary depending on their background and tastes.

- **Security architecture:** A unified security design that addresses the necessities and potential risks involved in a certain scenario or environment. It also specifies when and where to apply security controls. The design process is generally reproducible ([Stallings, 2018](#)).

- **Security operations:** The process that identifies critical information to determine if friendly actions can be observed by enemy intelligence and if information obtained by adversaries could be useful to them, and then executes selected measures that eliminate or reduce adversary exploitation of friendly critical information ([Sharp, 2006](#)).
- **Governance:** The framework for managing performance and risk; oversight of compliance and control responsibilities; and defining the cyber mission by mapping the structure, authority, and processes to create an effective program ([Blue Lance, n.d.](#)).
- **Physical security:** The protection of personnel, hardware, software, networks, and data from physical actions and events that could cause serious loss or damage to an enterprise, agency, or institution. This includes protection from fire, flood, natural disasters, burglary, theft, vandalism, and terrorism ([Healthcare Information and Management Systems Society, 2013](#)).
- **Threat intelligence:** Evidence-based knowledge, including context, mechanisms, indicators, implications, and actionable advice, about an existing or emerging threat to assets, which can be used to inform decisions about responding to that threat ([McMillan, 2013](#)).
- **Career development:** Training of future cybersecurity professionals.
- **Risk assessment:** Analyzing what can go wrong, how likely it is to happen, what the potential consequences are, and how tolerable the identified risk is.
- **User education:** Teaching users how to protect themselves from cyberattacks by informing them of risks, exploits, and external threats, as well as teaching them the skills needed to combat common attacks.
- **Frameworks and standards:** The creation of new security frameworks and practices for professionals to follow.
- **Supervisory Control and Data Acquisition (SCADA):** Protecting and securing large scale infrastructure components (like the electric grid, water treatment systems, and factories) from cyberattacks.

Our program is primarily concerned with the technical aspects of cybersecurity, but we will connect these areas to the overall security organization and business.

- We'll dive deep into security operations and security architecture in this course. We'll also explore penetration testing and discuss red team vs. blue team in depth.
- We'll consider the importance of user education and training. We'll discuss physical security tools like key cards, and the implementation of disaster recovery planning.
- Throughout this course, we look at different frameworks and standards. It is important to understand how these affect the development and priorities of a

security organization. It's also important to note that different industries follow different frameworks, which we'll discuss next week.

04. Activity: Cybersecurity Careers Research

[M1.3 Activity File: Careers](#)

05. Facilitated Discussion: Cybersecurity Careers Research

It's important for you to assess your existing skills and job level:

- Are you a career switcher or starter? Are you entering the security field for the first time?
- If you have existing experience, what is it? Possibilities include:
 - Technical cybersecurity adjacent fields, such as IT and networking;
 - Technical experience, such as software development; and
 - Security experience, such as security operations center (SOC) operations, security analysis, and security risk analysis and policy.

Similar job titles can have different skills and qualifications, so it's important to review the specifics of a job description.

- You'll notice that many jobs, such as the role of a security analyst, don't necessarily expect candidates to have a lot of specific skills and tools. Many companies are open to teaching employees on the job. What's important is showing your passion, ability, and willingness to learn.
- Passion and ability to learn is key. This program will push you past your comfort zone, but learning how to learn is a critical skill that employers search for.
 - Also, always stay curious and keep up to date on cybersecurity news and trends. Complete projects and participate in activities like Capture the Flag (CTF) competitions. In interviews, you should be able to discuss and demonstrate your passion and how you learn.
- Do not be deterred by jobs that require a few more years of experience than what you have. If you are just starting out, a job that requires six to eight years of

experience is most likely not a possibility, but a job that requires two years of experience is often possible.

- Employers will often intentionally put in more years of experience than are actually needed in order to avoid receiving too many applications.
- According to the [US Bureau of Labor Statistics](#) (2023), there will be a 35% growth in employment for information security analysts between the years 2021 and 2031.
- The 2022 State of Cybersecurity report by [ISACA](#) states:
 - 63% of respondent enterprises have unfilled cybersecurity positions.
 - 60% of respondent enterprises report experiencing difficulties in retaining qualified cybersecurity professionals.
 - 82% of respondents believe that leadership sees value in conducting cyber-risk assessments (but only 41% perform annual assessments).
- Also, according to [ISC2](#) (2018), 87% of all cybersecurity professionals started their careers in a different field. This means that the majority of cybersecurity professionals have backgrounds similar to yours.

While it's possible to jump between professional levels, in general, the cybersecurity industry wants candidates with proven experience when hiring. If you're new to the field, you will need to put in work to get started.

- However, it's important to note that because of the increased demand for security professionals, you can probably skip some of the years of required experience for certain jobs. We have had complete beginners in our program get jobs as network administrators, SOC analysts, and security analysts.

Now we will discuss common job titles and where they fit in different pathways.

- IT support specialists, computer help desk specialists, and other similar titles are entry-level jobs for all cybersecurity domains.
- Systems administrator (sysadmin) and network administrator jobs are also entry-level jobs in the networking space.
 - Technical cybersecurity roles require a strong foundation in networking, which our program covers. We also provide relevant cybersecurity context to our networking modules, so they are still relevant for those who already have sysadmin or networking experience. These roles are within reach for career switchers and starters.
 - Networking roles focus on configuring and maintaining networks, and often include security-related tasks such as monitoring logs, implementing firewalls, and maintaining network-based backups.

- A range of entry-level security jobs focus on the operations and defensive side of security. Our program will equip you with the skills to be hired in these roles, like many of our graduates.
 - SOC analysts are the most common entry-level security jobs. Specifically, Tier 1 SOC analysts can be hired with no prior professional experience. They work at an organization's SOC, monitoring and analyzing network traffic.
 - They are the first level of defense and are responsible for monitoring the security of users, systems, and the network. They detect and investigate suspicious and potentially malicious activity on the network and within systems.
 - Security or information specialists or analysts encompass different skill levels and responsibilities, so it is important to always review the job description, specific skills, and qualifications. These roles actively protect an organization from security threats and breaches.
 - Common responsibilities for entry-level roles include keeping up to date with security improvements, monitoring and protecting security systems, implementing security tools and protocols, developing security requirements, analyzing security policies and protocols for weaknesses and flaws, and designing and developing training.
 - Information assurance technicians protect computer assets by establishing and documenting access and maintaining files.
 - Forensics specialists come in after a crime has been committed. They analyze logs and other data to determine what occurred.
 - Some common responsibilities include conducting security incident investigations, recovering and examining data from computers and other devices, compiling evidence for legal cases, writing technical reports, preparing evidence for trial, and working with law enforcement on the validity and preservation of data.
 - Risk analysts use their knowledge and experience to examine systems and procedures to identify potential adverse events, including hardware and software crashes, physical disasters, malicious intruders, malware, denial-of-service attacks, and employee misconduct.
 - IT auditors are responsible for the internal controls and risks to a company's technology network. This role includes identifying the weaknesses in a system's network and creating an action plan to prevent security breaches in the technology.
- Typically, with some years of experience in the above roles, people can move on to mid-level or advanced-level roles, such as the ones that follow. Many mid-level and advanced-level roles may also require additional certifications.

- Security engineers test the network for vulnerabilities, monitor for security breaches, and develop security plans and policies.
 - Common responsibilities include installing and configuring firewalls and intrusion detection systems, performing vulnerability testing and security assessments, evaluating new technologies, testing existing security solutions, and defining, implementing, and maintaining security policies.
 - Many of these responsibilities are similar to those of a security analyst, but are more advanced and often more technical.
- Penetration testers or ethical hackers are often hired by a company to test and break into their security controls and defenses, and identify and report vulnerabilities and flaws in a system, using specific tools to simulate cyberattacks.
 - Their goal is to help companies improve their security posture. Pentesters need to deeply understand the types of systems they're trying to break into. They usually specialize in specific systems, such as networks, web applications, and mobile applications.
- Chief information officers (CIO) and chief information security officers (CISO) are advanced managerial roles in cybersecurity. Certifications such as CISSP+ are often required.
- This is just the start. New jobs are constantly emerging to take advantage of new technologies such as machine learning and artificial intelligence.

Additionally, there are new jobs being created regularly. Some examples of emerging cybersecurity jobs include:

- IoT security specialist: As the Internet of Things (IoT) continues to expand, the need for specialists who can secure connected devices is increasing. IoT security specialists focus on protecting networks that include smart devices in homes, factories, and cities from attacks and vulnerabilities.
- AI security specialist: With the integration of AI into various business processes and consumer applications, securing AI systems has become critical. AI security specialists develop strategies to protect AI systems from malicious attacks and ensure that AI operates as intended without bias.
- Cloud security architect: As more organizations move their operations to the cloud, the demand for cloud security architects who can design secure cloud environments and strategies has skyrocketed. These professionals are responsible for ensuring that the cloud services used by a business are secure from data breaches and meet compliance requirements.

- Threat hunter: Threat hunters proactively search for malware or attackers that are lurking undetected in a network. Unlike other security roles that are more reactive, threat hunters use knowledge of the latest hacking techniques to predict and identify potential threats before they cause harm.
- Cybersecurity incident response specialist: This role involves handling security breaches and attacks when they occur. Cybersecurity incident response specialists are tasked with quickly containing the incident, mitigating any damage, and ensuring that the organization can recover as swiftly as possible. They also perform post-incident analysis to prevent future threats.

Can you think of any other unique or unusual cybersecurity jobs?

06. Using Our Career Services Department

Our Career Services department is available to you from the first day of the program.

- During class, you will tie concepts and skills in with career outcomes and roles. The Career Services team will help you do the practical, hard work of obtaining a new role or promotion.
 - There are several career milestones for you to complete that you'll find in Canvas.
 - You can access them all now and submit milestones whenever you want during the program.
 - However, you will need to submit at least one milestone in order to unlock Career Services.
 - One of the milestones, an updated, polished resume, will be a requirement in one of our later Challenge assignments, but you can submit this earlier if you'd like.
 - Review the slides on Becoming Employer Competitive, Working with Your Profile Coach, and Working with Your Career Director.
 - This week's Challenge will contain more details on Career Services. If you do not know who your Career Director is, reach out to your Student Success Manager.
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07. Break

08. Cybersecurity Certifications

As the demand for cybersecurity careers grows, employers frequently use certification as a measure of employee qualifications and training when hiring.

- If you're serious about advancing your career in security, certifications will be an important step toward more opportunity, knowledge, and skills. Certifications can be used across jobs and organizations.
- Note that it is possible to get an entry-level security job without any certifications. For example, an SOC analyst is a common job that often does not require any certifications. For career starters, many networking jobs, such as systems or network administrators, also do not require certifications.

Certifications are obtained through accrediting organizations. Here is a list of top providers, with a few of the certifications they offer:

- CompTIA: Security+, PenTest+
- GIAC: GPEN, GCIH
- ISC2: CISSP, SSCP
- Offensive Security: OSCP, OSWP

Most of these organizations offer both introductory certifications and more advanced specialization certifications.

- As of early 2024, there are over 400 security certifications available to cybersecurity professionals.
- Many of these certifications focus on specific vendors, industries, and government and industry compliance.
- Review the following list of certifications:
 - [Wikipedia: Computer Security Certifications](#)
- Another excellent resource is [Paul Jeremy's Security Certification Roadmap](#).
 - This roadmap breaks down cybersecurity into multiple fields, ranging from communication and network security to security operations. It also indicates the relative difficulty of the exams. Clicking on any certification will take you to the home page of the certification.
- You can refer back to this tool as you develop your career interests throughout the class.
- Today's class will focus only on the Security+ exam. Additionally, many of the topics in this course will help prepare you for this exam.

09. Introduction to Security+ Certification

While this class explores the importance of cybersecurity certifications, this course is not a test prep course. This course will offer an exam voucher and practice resources for the Security+ exam, but to be properly prepared to take the exam, you will have to learn other topics and practice extensively on your own.

What is the Security+ certification?

- According to CompTIA: "Security+ is the first security certification IT professionals should earn. It establishes the core knowledge required of any cybersecurity role and provides a springboard to intermediate-level cybersecurity jobs. Security+ incorporates best practices in hands-on troubleshooting to ensure security professionals have practical security problem-solving skills. Cybersecurity professionals with Security+ know how to address security incidents—not just identify them."

What are some jobs that may require the Security+ certification?

- Security architect
- Security engineer
- Security consultant
- Security specialist
- Information security analyst

As of January 2024, the average annual pay for an information security analyst in the US was \$96,652 per year ([ZipRecruiter, n.d.](#)).

What skills and knowledge areas are assessed in this certification?

Note the following domains covered in the exam:

- **Attacks, threats, and vulnerabilities:** Covers the identification of various types of security threats and vulnerabilities, including the tactics, techniques, and procedures used by attackers.
- **Implementation:** Deals with the practical application of security concepts, involving the installation and configuration of security controls and technologies.
- **Architecture and design:** Focuses on the principles of secure network design and architecture, including cloud, mobile, and IoT environments.

- **Operations and incident response:** Explores how to manage and protect resources using appropriate security measures, and how to respond to and recover from security incidents.
- **Governance, risk, and compliance:** Addresses the management of cybersecurity risk and compliance with security policies and regulations, ensuring organizational policies meet legal and regulatory requirements.

We will cover many aspects of these topics throughout the program. Throughout the course, we will emphasize which topics and skills covered in class are also relevant to the Security+ exam.

- You will still need to study topics in depth outside of class to prepare for the certification.

At which point in my career should I consider taking the Security+ exam?

- Security+ is considered an entry-level exam. The skills that you learn in this program will provide a foundation that you can build on to prepare for the exam.
- In class, we will do lots of practical work and learn about several topics in depth, but the Security+ exam is more broad and requires knowledge in areas that we do not cover in this program.
- For these gaps, you will get access to a question-based tool called CertMaster Practice, which will provide the necessary coverage needed to master the exam.

Why are all the topics in the Security+ exam not covered in our program?

Again, this boot camp is not a test prep course.

- We don't cover every topic discussed in the Security+ exam because our program focuses on providing hands-on experience with the most common concepts, tools, and technologies used in security and networking.
- What the course does cover provides you with knowledge that will be relevant to almost any security job you'll apply for, as well as a framework that makes it easy to learn new protocols and tools when you encounter them in your studies and work.
- Some topics on the Security+ exam are left out because they are highly specific and relevant to either only the test, or to specific subfields of cybersecurity. As such, they contribute little to foundational, general knowledge.
 - For example, the TACACS+ protocol appears on the Security+ exam, but only engineers who work specifically with Cisco devices use it.

- These topics are left out so you can spend the majority of the course learning the tools and concepts that will prepare you to succeed in the largest number of jobs, and to learn any networking or security concepts you encounter.
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10. Introduction to Security+ Certification Exam

- The test consists of 90 multiple choice and performance-based questions.
- Test takers will have 90 minutes to complete the exam.
- The passing score is 750 (on a scale of 100–900).
- Students who pass the boot camp will receive an exam voucher at the end of the program.

In a later module, we will go into the type of questions on the test and specific strategies to complete them.

11. Challenge Introduction

This week's Challenge has two parts:

1. If you haven't done so, you should complete the Digital Literacy Assessment in the prework. This will help identify any skills and knowledge gaps, so you can get the necessary support.
2. Most security professionals use prominent reports, blogs, and papers to stay up to date on the field. Since this is such a critical skill, this week's Challenge requires you to review three prominent security reports in order to get a basic understanding of the market. As you review these reports you will be asked to answer a series of questions.

If you have questions about due dates, consult Canvas.

12. Fireside Chat

We have covered a lot in the first week of the program. For the rest of class, we will open the floor to your questions and discuss how your instructor and TAs got into cybersecurity.

13. Wrap-Up

If you feel overwhelmed by the amount of information covered, don't worry. You will become familiar with these concepts throughout the course. After you receive the CertMaster Practice tool, you will have a full year of access to the platform and can self-study topics at your own pace.

- While we did not do a deep dive into the other domains, we will touch on them throughout the modules in this program, and we will help you understand what to study for each.